

A COUNTEREXAMPLE TO A CARTESIAN-PRODUCT CONJECTURE FOR NODE RESISTANCE CURVATURE

ABSTRACT. We disprove Conjecture 4 of Dawkins et al. as stated in arXiv:2403.01037v1. Two rooted trees on 5 and 8 vertices have root curvature zero, whereas the corresponding vertex in their Cartesian product has node resistance curvature $32552200/4351396379 > 0$. The verification uses exact integer arithmetic and the matrix-tree theorem.

1. THE COUNTEREXAMPLE

Let $G = (V, E)$ be a finite connected simple graph with unit edge conductances, let L_G be its Laplacian, and let L_G^+ be the Moore–Penrose pseudoinverse of L_G . For $u, v \in V$, write

$$\omega_G(u, v) = (\mathbf{e}_u - \mathbf{e}_v)^\top L_G^+ (\mathbf{e}_u - \mathbf{e}_v)$$

for the effective resistance. The node resistance curvature introduced by Devriendt and Lambiotte [2] is

$$p_v(G) = 1 - \frac{1}{2} \sum_{u \sim v} \omega_G(u, v).$$

We write $G_1 \square G_2$ for the Cartesian product.

In arXiv:2403.01037v1, Dawkins et al. [1] state the following.

Conjecture 4. *For graphs G_1, G_2 and vertices $i \in V(G_1)$ and $j \in V(G_2)$, if*

$$p_i(G_1) \leq 0 \quad \text{and} \quad p_j(G_2) \leq 0,$$

then

$$p_{(i,j)}(G_1 \square G_2) < 0.$$

Theorem 1. *Let G_1 have vertex set $\{0, 1, 2, 3, 4\}$ and edge set*

$$E(G_1) = \{01, 02, 13, 34\},$$

and let G_2 have vertex set $\{0, 1, \dots, 7\}$ and edge set

$$E(G_2) = \{01, 02, 03, 04, 15, 56, 57\},$$

where ab denotes the edge $\{a, b\}$. At the marked vertices $i = j = 1$,

$$p_i(G_1) = p_j(G_2) = 0,$$

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but

$$p_{(i,j)}(G_1 \square G_2) = \frac{32552200}{4351396379} > 0.$$

Consequently, Conjecture 4 is false as stated.

Proof. Both G_1 and G_2 are trees, and the marked vertex has degree 2 in each. Every unit edge of a tree has effective resistance 1, so

$$p_i(G_1) = p_j(G_2) = 1 - \frac{2}{2} = 0.$$

Set $H = G_1 \square G_2$ and $x = (1, 1)$. The neighbors of x are

$$(0, 1), \quad (3, 1), \quad (1, 0), \quad (1, 5).$$

For a connected graph K , let $\tau(K)$ denote its number of spanning trees. If $e = uv$ is a unit edge, then

$$\omega_K(u, v) = \Pr(e \in \mathbf{T}) = \frac{\tau(K/e)}{\tau(K)},$$

where $\mathbf{T}$ is a uniformly random spanning tree and K/e is obtained by contracting e , with parallel edges retained with multiplicity. The first equality is the spanning-tree inclusion identity [2, Theorem 2], and the second follows from the standard contraction bijection.

Applying the matrix-tree theorem to the reduced Laplacians gives

$$\tau(H) = 723,963,572,556,125$$

and the following four cofactors.

contracted edge e	$\tau(H/e)$
$\{x, (0, 1)\}$	359,244,778,688,125
$\{x, (3, 1)\}$	359,244,778,688,125
$\{x, (1, 0)\}$	354,728,028,191,625
$\{x, (1, 5)\}$	363,877,814,994,375

Therefore

$$\begin{aligned} p_x(H) &= \frac{2\tau(H) - \sum_{e \ni x} \tau(H/e)}{2\tau(H)} \\ &= \frac{10,831,744,550,000}{1,447,927,145,112,250} \\ &= \frac{32552200}{4351396379} > 0. \end{aligned}$$

This contradicts the asserted strict negativity. $\square$

Remark. The counterexample uses equality in both hypotheses. It therefore refutes Conjecture 4 as stated, but it does not address the variant with $p_i(G_1) < 0$ and $p_j(G_2) < 0$. No minimality claim is made.

REFERENCES

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